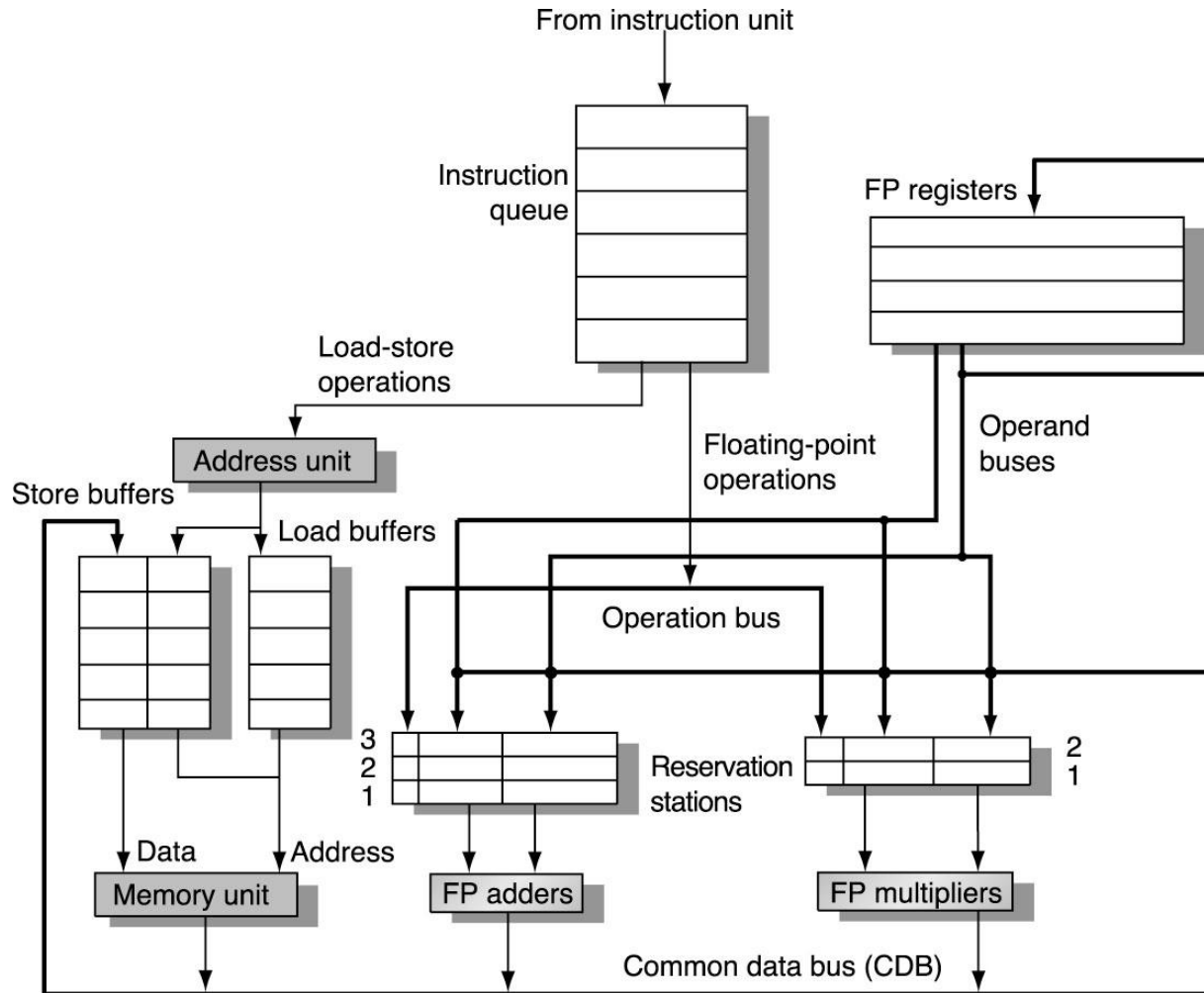


Examples of behavior of the Tomasulo's architecture

Tomasulo Organization



Reservation Station Components

Op—Operation to perform in the unit (e.g., + or −)

Qj, Qk—Reservation stations producing source registers

Vj, Vk—Value of Source operands

Rj, Rk—Flags indicating when Vj, Vk are ready

Busy—Indicates reservation station and FU is busy

Register result status—Indicates which functional unit will write each register, if one exists. Blank when no pending instructions that will write that register.

Three Stages of Tomasulo Algorithm

1. **Issue**—get instruction from FP Op Queue

If reservation station free, the scoreboard issues instr & sends operands (renames registers).

2. **Execution**—operate on operands (EX)

When both operands ready then execute;
if not ready, watch CDB for result

3. **Write result**—finish execution (WB)

Write on Common Data Bus to all awaiting units;
mark reservation station available.

Tomasulo Example Cycle 0

Instruction Status				Issue	Execution Complete	Write Result						
Instruction		j	k									
LD	F6	34+	R2					Load1	No			
LD	F2	45+	R3					Load2	No			
MULTD	F0	F2	F4					Load3	No			
SUBD	F8	F6	F2									
DIVD	F10	F0	F6									
ADDD	F6	F8	F2									
Reasevation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
		0 Add1	No									
		0 Add2	No									
		0 Add3	No									
		0 Mult1	No									
		0 Mult2	No									
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
0			FU									

Tomasulo Example Cycle 1

Instruction Status				Issue	Execution Complete	Write Result						
Instruction		j	k									
LD	F6	34+	R2	1				Load1	Yes	34+R2		
LD	F2	45+	R3					Load2	No			
MULTD	F0	F2	F4					Load3	No			
SUBD	F8	F6	F2									
DIVD	F10	F0	F6									
ADDD	F6	F8	F2									
Resevation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	0	Mult1	No									
	0	Mult2	No									
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
1			FU				Load1					

Tomasulo Example Cycle 2

Instruction Status				Issue	Execution Complete	Write Result						
Instruction		j	k									
LD	F6	34+	R2	1				Load1	Yes	34+R2		
LD	F2	45+	R3	2				Load2	Yes	45+R3		
MULTD	F0	F2	F4					Load3	No			
SUBD	F8	F6	F2									
DIVD	F10	F0	F6									
ADDD	F6	F8	F2									
Resevation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	0	Mult1	No									
	0	Mult2	No									
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
2			FU		Load2		Load1					

Tomasulo Example Cycle 3

Instruction Status				Issue	Execution Complete	Write Result						
Instruction		j	k									
LD	F6	34+	R2	1	3			Load1	Yes	34+R2		
LD	F2	45+	R3	2				Load2	Yes	45+R3		
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2									
DIVD	F10	F0	F6									
ADDD	F6	F8	F2									
Resevation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	0	Mult1	Yes	MULTD		R[F4]	Load2					
	0	Mult2	No									
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
3			FU	Mult1	Load2		Load1					

Tomasulo Example Cycle 4

Instruction Status				Issue	Execution Complete	Write Result						
Instruction		j	k									
LD	F6	34+	R2	1	3	4			Load1	No		
LD	F2	45+	R3	2	4				Load2	Yes	45+R3	
MULTD	F0	F2	F4	3					Load3	No		
SUBD	F8	F6	F2	4								
DIVD	F10	F0	F6									
ADDD	F6	F8	F2									
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	Yes	SUBD	M[34+R2]			Load2				
	0	Add2	No									
	0	Add3	No									
	0	Mult1	Yes	MULTD		R[F4]	Load2					
	0	Mult2	No									
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
4			FU	Mult1	Load2		M[34+R2]	Add1				

Tomasulo Example Cycle 5

Instruction Status				Execution		Write						
Instruction		j	k	Issue	Complete	Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4								
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2									
Reservation Stations						S1	S2	RS for j	RS for k			
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	Yes	SUBD	M[34+R2]	M[45+R3]						
	0	Add2	No									
	0	Add3	No									
	0	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
5			FU	Mult1	M[45+R3]		M[34+R2]	Add1	Mult2			

Tomasulo Example Cycle 6

<u>Instruction Status</u>				Issue	Execution Complete	Write Result			Busy	Address		
Instruction		j	k									
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4								
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6								
<u>Reservation Stations</u>					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	2	Add1	Yes	SUBD	M[34+R2]	M[45+R3]						
	0	Add2	Yes	ADDD		M[45+R3]	Add1					
	0	Add3	No									
	10	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
<u>Register result status</u>												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
6			FU	Mult1	M[45+R3]		Add2	Add1	Mult2			

Tomasulo Example Cycle 7

<u>Instruction Status</u>				Issue	Execution Complete	Write Result			Busy	Address		
Instruction		j	k									
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7							
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6								
<u>Reservation Stations</u>					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	1	Add1	Yes	SUBD	M[34+R2]	M[45+R3]						
	0	Add2	Yes	ADDD		M[45+R3]	Add1					
	0	Add3	No									
	9	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
<u>Register result status</u>												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
7			FU	Mult1	M[45+R3]		Add2	Add1	Mult2			

Tomasulo Example Cycle 8

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6								
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	Yes	ADDD	M[]-M[]	M[45+R3]						
	0	Add3	No									
	8	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
8			FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 9

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6								
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	2	Add2	Yes	ADDD	M[]-M[]	M[45+R3]						
	0	Add3	No									
	7	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
9			FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 10

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10							
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	1	Add2	Yes	ADDD	M[]-M[]	M[45+R3]						
	0	Add3	No									
	6	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
10			FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 11

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	5	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
11			FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 12

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	4	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
12			FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 13

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	3	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
13			FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 14

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3				Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	2	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
14			FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 15

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3	15			Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	1	Mult1	Yes	MULTD	M[45+R3]	R[F4]						
	0	Mult2	Yes	DIVD		M[34+R2]	Mult1					
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
15			FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 16

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3	15	16		Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
		0 Add1	No									
		0 Add2	No									
		0 Add3	No									
		0 Mult1	No									
		0 Mult2	Yes	DIVD	M*F4	M[34+R2]						
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
16			FU	M*F4	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 17

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3	15	16		Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	0	Mult1	No									
	40	Mult2	Yes	DIVD	M*F4	M[34+R2]						
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
17		FU		M*F4	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 18

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3	15	16		Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5								
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	0	Mult1	No									
	39	Mult2	Yes	DIVD	M*F4	M[34+R2]						
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
18		FU		M*F4	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 56

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3	15	16		Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5	56							
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
	0	Add1	No									
	0	Add2	No									
	0	Add3	No									
	0	Mult1	No									
	1	Mult2	Yes	DIVD	M*F4	M[34+R2]						
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
56		FU		M*F4	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 57

Instruction Status		j	k	Issue	Execution Complete	Write Result			Busy	Address		
LD	F6	34+	R2	1	3	4		Load1	No			
LD	F2	45+	R3	2	4	5		Load2	No			
MULTD	F0	F2	F4	3	15	16		Load3	No			
SUBD	F8	F6	F2	4	7	8						
DIVD	F10	F0	F6	5	56	57						
ADDD	F6	F8	F2	6	10	11						
Reservation Stations					S1	S2	RS for j	RS for k				
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk				
		0 Add1	No									
		0 Add2	No									
		0 Add3	No									
		0 Mult1	No									
		0 Mult2	No									
Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
57		FU		M*F4	M[45+R3]		(M-M)+M	M[]-M[]	M*F4/M			

Tomasulo Loop Example

Loop: LD	F0	0	R1
MULTD	F4	F0	F2
SD	F4	0	R1
SUBI	R1	R1	#8
BNEZ	R1	Loop	

- Multiply takes 4 clock cycles
- Loads have cache misses

Loop Example Cycle 0

Instruction status				ExecutionWrite			Busy	Address					
Instruction	j	k	iteration	Issue	complete	Result							
LD F0	0	R1	1			Load1	No	Qi					
MULT F4	F0	F2	1			Load2	No						
SD F4	0	R1	1			Load3	No						
LD F0	0	R1	2			Store1	No						
MULT F4	F0	F2	2			Store2	No						
SD F4	0	R1	2			Store3	No						
Reservation Stations				S1	S2	RS for j	RS for k	Code:					
Time	Name	Busy	Op	Vj	Vk	Qj	Qk						
0	Add1	No						LD F0 0 R1					
0	Add2	No						MULT F4 F0 F2					
0	Add3	No						SD F4 0 R1					
0	Mult1	No						SUBI R1 R1 #8					
0	Mult2	No						BNEZ R1 Loop					
Register result status													
Clock	R1		F0	F2	F4	F6	F8	F10	F12	...	F30		
0	80	Qi											

Loop Example Cycle 1

Instruction status				ExecutionWrite			Busy	Address			
Instruction	j	k	iteration	Issue	complete	Result					
LD F0	0	R1	1	1		Load1	Yes	80			
MULT F4	F0	F2	1			Load2	No				
SD F4	0	R1	1			Load3	No	Qi			
LD F0	0	R1	2			Store1	No				
MULT F4	F0	F2	2			Store2	No				
SD F4	0	R1	2			Store3	No				
Reservation Stations				S1	S2	RS for j	RS for k	Code:			
Time	Name	Busy	Op	Vj	Vk	Qj	Qk				
0	Add1	No						LD F0 0 R1			
0	Add2	No						MULT F4 F0 F2			
0	Add3	No						SD F4 0 R1			
0	Mult1	No						SUBI R1 R1 #8			
0	Mult2	No						BNEZ R1 Loop			
Register result status											
Clock	R1		F0	F2	F4	F6	F8	F10	F12...	F30	
1	80	Qi	Load1								

Loop Example Cycle 2

Instruction status				ExecutionWrite			Busy	Address
Instruction	j	k	iteration	Issue	complete	Result		
LD F0	0	R1	1	1		Load1	Yes	80
MULTD F4	F0	F2	1	2		Load2	No	
SD F4	0	R1	1			Load3	No	Qi
LD F0	0	R1	2			Store1	No	
MULTD F4	F0	F2	2			Store2	No	
SD F4	0	R1	2			Store3	No	

Reservation Stations			S1	S2	RS for j	RS for k	Code:
Time	Name	Busy Op	Vj	Vk	Qj	Qk	
0	Add1	No					LD F0 0 R1
0	Add2	No					MULTD F4 F0 F2
0	Add3	No					SD F4 0 R1
0	Mult1	Yes MULTD		R(F2)	Load1		SUBI R1 R1 #8
0	Mult2	No					BNEZ R1 Loop

Register result status		F0	F2	F4	F6	F8	F10	F12...	F30
Clock	R1								
2	80 Qi	Load1		Mult1					

Loop Example Cycle 3

Instruction status				ExecutionWrite			Busy	Address
Instruction	j	k	iteration	Issue	complete	Result		
LD F0	0	R1	1	1		Load1	Yes	80
MULTD F4	F0	F2	1	2		Load2	No	
SD F4	0	R1	1	3		Load3	No	Qi
LD F0	0	R1	2			Store1	Yes	80
MULTD F4	F0	F2	2			Store2	No	
SD F4	0	R1	2			Store3	No	

Reservation Stations			S1	S2	RS for j	RS for k	Code:
Time	Name	Busy Op	Vj	Vk	Qj	Qk	
0	Add1	No					LD F0 0 R1
0	Add2	No					MULTD F4 F0 F2
0	Add3	No					SD F4 0 R1
0	Mult1	Yes MULTD		R(F2)	Load1		SUBI R1 R1 #8
0	Mult2	No					BNEZ R1 Loop

Register result status		F0	F2	F4	F6	F8	F10	F12...	F30
Clock	R1								
3	80 Qi	Load1		Mult1					

Loop Example Cycle 4

Instruction status					Execution	Write					
Instruction	<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>		Busy	Address		
LD F0	0	R1	1	1			Load1	Yes	80		
MUL F4	F0	F2	1	2			Load2	No			
SD F4	0	R1	1	3	4		Load3	No		Qi	
LD F0	0	R1	2				Store1	Yes	80	Mult1	
MUL F4	F0	F2	2				Store2	No			
SD F4	0	R1	2				Store3	No			
Reservation Stations				<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>				
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>		
	0	Add1	No						LD F0	0	R1
	0	Add2	No						MUL F4	F0	F2
	0	Add3	No						SD F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB R1	R1	#8
	0	Mult2	No						BNE R1	Loop	
Register result status											
Clock	R1		F0	F2	F4	F6	F8	F10	F12	...	F30
4	80	Qi	Load1		Mult1						

Loop Example Cycle 5

Instruction status					Executing	Write					
Instruction	<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>		Busy	Address		
LD F0	0	R1	1	1			Load1	Yes	80		
MUL F4	F0	F2	1	2			Load2	No			
SD F4	0	R1	1	3	4		Load3	No		Qi	
LD F0	0	R1	2				Store1	Yes	80	Mult1	
MUL F4	F0	F2	2				Store2	No			
SD F4	0	R1	2				Store3	No			
Reservation Stations				<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>				
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>		
	0	Add1	No						LD F0	0	R1
	0	Add2	No						MUL F4	F0	F2
	0	Add3	No						SD F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB R1	R1	#8
	0	Mult2	No						BNE R1	Loop	
Register result status											
Clock	R1		F0	F2	F4	F6	F8	F10	F12	...	F30
5	80	Qi	Load1		Mult1						

Loop Example Cycle 6

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1		Load1	Yes	80	
MUL	F4	F0	F2	1	2		Load2	Yes	72	
SD	F4	0	R1	1	3	4	Load3	No		Qi
LD	F0	0	R1	2	6		Store1	Yes	80	Mult1
MUL	F4	F0	F2	2			Store2	No		
SD	F4	0	R1	2			Store3	No		
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB	R1 R1 #8
	0	Mult2	No						BNE	R1 Loop
Register result status										
Clock	R1		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...
6	72	Qi	Load2		Mult1					

Loop Example Cycle 7

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1		Load1	Yes	80	
MUL	F4	F0	F2	1	2		Load2	Yes	72	
SD	F4	0	R1	1	3	4	Load3	No		Qi
LD	F0	0	R1	2	6		Store1	Yes	80	Mult1
MUL	F4	F0	F2	2	7		Store2	No		
SD	F4	0	R1	2			Store3	No		
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB	R1 R1 #8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1 Loop
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F12	... F30
7	72	Qi	Load2		Mult2					

Loop Example Cycle 7

	Instruction	<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address		
In	LD F0	0	R1	1	1			Load1	Yes	80	
LI	MUL F4	F0	F2	1	2			Load2	Yes	72	
MS	SD F4	0	R1	1	3	4		Load3	No		Qi
S	SUB R1	R1	#8	1	4	5	6				
LI	BNE R1	Loop		1	5	7	-				
M	LD F0	0	R1	2	6			Store1	Yes	80	Mult1
SI	MUL F4	F0	F2	2	7			Store2	No		
R	SD F4	0	R1	2				Store3	No		
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>			
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>		
	0	Add1	No						LD F0	0	R1
	0	Add2	No						MUL F4	F0	F2
	0	Add3	No						SD F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB R1	R1	#8
R	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE R1	Loop	
C	Register result status										
	Clock	R1		F0	F2	F4	F6	F8	F10	F12	... F30

Loop Example Cycle 8

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1		Load1	Yes	80	
MUL	F4	F0	F2	1	2		Load2	Yes	72	
SD	F4	0	R1	1	3	4	Load3	No		Qi
LD	F0	0	R1	2	6		Store1	Yes	80	Mult1
MUL	F4	F0	F2	2	7		Store2	Yes	72	Mult2
SD	F4	0	R1	2	8		Store3	No		
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB	R1 R1 #8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1 Loop
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F12	... F30
8	72	Qi	Load2		Mult2					

Loop Example Cycle 9

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>		Busy	Address	
LD	F0	0	R1	1	1	9		Load1	Yes	80	
MUL	F4	F0	F2	1	2			Load2	Yes	72	
SD	F4	0	R1	1	3	4		Load3	No		Qi
LD	F0	0	R1	2	6			Store1	Yes	80	Mult1
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2
SD	F4	0	R1	2	8	9		Store3	No		
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>			
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>		
	0	Add1	No						LD	F0	0 R1
	0	Add2	No						MUL	F4	F0 F2
	0	Add3	No						SD	F4	0 R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB	R1	R1 #8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1	Loop
Register result status											
Clock	R1		F0	F2	F4	F6	F8	F10	F12	...	F30
9	72	Qi	Load2		Mult2						

Loop Example Cycle 10

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2			Load2	Yes	72
SD	F4	0	R1	1	3	4		Load3	No	Qi
LD	F0	0	R1	2	6	10		Store1	Yes	80
MUL	F4	F0	F2	2	7			Store2	Yes	72
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	Yes	MULTD	M(80)	R(F2)			SUB	R1 R1 #8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1 Loop
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F11	... F30
10	72	Qi	Load2		Mult2					

Loop Example Cycle 11

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2			Load2	No	
SD	F4	0	R1	1	3	4		Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 Mult1
MUL	F4	F0	F2	2	7			Store2	Yes	72 Mult2
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	4	Mult1	Yes	MULTD	M(80)	R(F2)			SUB	R1 R1 #8
	0	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1 Loop
Register result status										
Clock		R1		F0	F2	F4	F6	F8	F10	F11... F30
11		64	Qi			Mult2				

Loop Example Cycle 12

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2			Load2	No	
SD	F4	0	R1	1	3	4		Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 Mult1
MUL	F4	F0	F2	2	7			Store2	Yes	72 Mult2
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	3	Mult1	Yes	MULTD	M(80)	R(F2)			SUB	R1 R1 #8
	4	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1 Loop
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F12	... F30
12	64	Qi			Mult2					

Loop Example Cycle 13

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2			Load2	No	
SD	F4	0	R1	1	3	4		Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 Mult1
MUL	F4	F0	F2	2	7			Store2	Yes	72 Mult2
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	2	Mult1	Yes	MULTD	M(80)	R(F2)			SUB	R1 R1 #8
	3	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1 Loop
Register result status										
Clock		R1		F0	F2	F4	F6	F8	F10	F11... F30
13		64	Qi			Mult2				

Loop Example Cycle 14

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2	14		Load2	No	
SD	F4	0	R1	1	3	4		Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 Mult1
MUL	F4	F0	F2	2	7			Store2	Yes	72 Mult2
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD F0 0 R1	
	0	Add2	No						MUL F4 F0 F2	
	0	Add3	No						SD F4 0 R1	
	1	Mult1	Yes	MULTD	M(80)	R(F2)			SUB R1 R1 #8	
	2	Mult2	Yes	MULTD	M(72)	R(F2)			BNE R1 Loop	
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F12...	F30
14	64	Qi			Mult2					

Loop Example Cycle 15

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2	14	15	Load2	No	
SD	F4	0	R1	1	3	4		Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 M(80)*R
MUL	F4	F0	F2	2	7	15		Store2	Yes	72 Mult2
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	No						SUB	R1 R1 #8
	1	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1 Loop
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F12	... F30
15	64	Qi			Mult2					

Loop Example Cycle 16

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2	14	15	Load2	No	
SD	F4	0	R1	1	3	4	16	Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 M(80)*R
MUL	F4	F0	F2	2	7	15	16	Store2	Yes	72 M(72)*R
SD	F4	0	R1	2	8	9		Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	Yes	MULTD		R(F2)	Load3		SUB	R1 R1 #8
	0	Mult2	No						BNE	R1 Loop
Register result status										
Clock	R1		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	<i>F14</i>
16	64	Qi			Mult1					

Loop Example Cycle 17

Instruction		<i>j</i>	<i>k</i>	<i>iteration</i>	<i>Issue</i>	<i>complete</i>	<i>Result</i>	Busy	Address	
LD	F0	0	R1	1	1	9	10	Load1	No	
MUL	F4	F0	F2	1	2	14	15	Load2	No	
SD	F4	0	R1	1	3	4	16	Load3	Yes	64 Qi
LD	F0	0	R1	2	6	10	11	Store1	Yes	80 M(80)*R
MUL	F4	F0	F2	2	7	15	16	Store2	Yes	72 M(72)*R
SD	F4	0	R1	2	8	9	17	Store3	No	
Reservation Stations					<i>S1</i>	<i>S2</i>	<i>RS for j</i>	<i>RS for k</i>		
	<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>	<i>Code:</i>	
	0	Add1	No						LD	F0 0 R1
	0	Add2	No						MUL	F4 F0 F2
	0	Add3	No						SD	F4 0 R1
	0	Mult1	Yes	MULTD		R(F2)	Load3		SUB	R1 R1 #8
	0	Mult2	No						BNE	R1 Loop
Register result status										
Clock	R1		F0	F2	F4	F6	F8	F10	F12	... F30
17	64	Qi			Mult1					